

SUBMITTED BY :

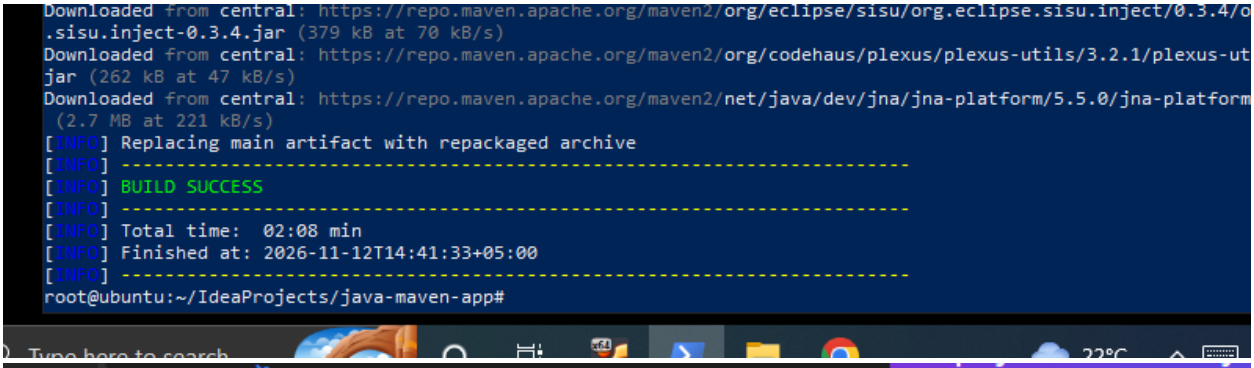
MALAYKA IBRAR

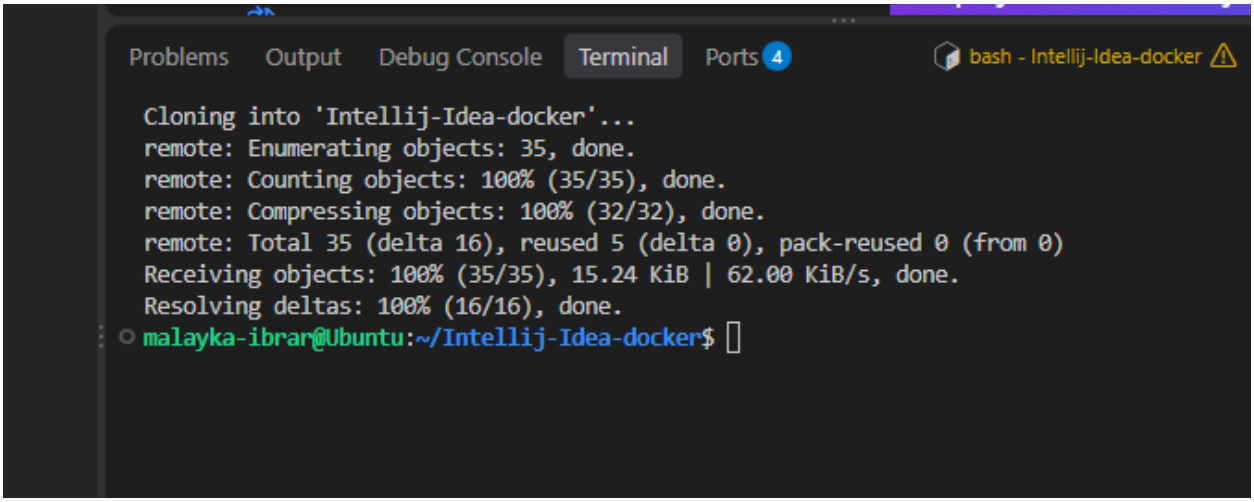
REGISTRATION NO:

058

Lab 5: Build and Run Maven, Gradle, and Node.js Projects in IntelliJ IDEA







```
root@ubuntu:~/IdeaProjects/java-maven-app# ls -l /root/.m2/settings.xml
-rw------- 1 root root 205 Nov 12 14:49 /root/.m2/settings.xml
root@ubuntu:~/IdeaProjects/java-maven-app# mvn deploy
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.example:java-maven-app >---------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
```

```
na-platform/5.5.0/jna-platform-5.5.0.jar (2.7 MB at 64 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/com/fasterxml/jack
son/core/jackson-databind/2.11.3/jackson-databind-2.11.3.jar (1.4 MB at 30 kB/s)
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  04:54 min
[INFO] Finished at: 2026-11-10T17:42:29+05:00
[INFO] ------------------------------------------------------------------------
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$
```

Terminal - @ubuntu:~/IdeaProjects/java-maven-app

File Edit View Terminal Tabs Help

```
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i
.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/s
rc/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1
.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.948 s
[INFO] Finished at: 2026-11-10T17:48:56+05:00
[INFO] ------------------------------------------------------------------------
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$
```

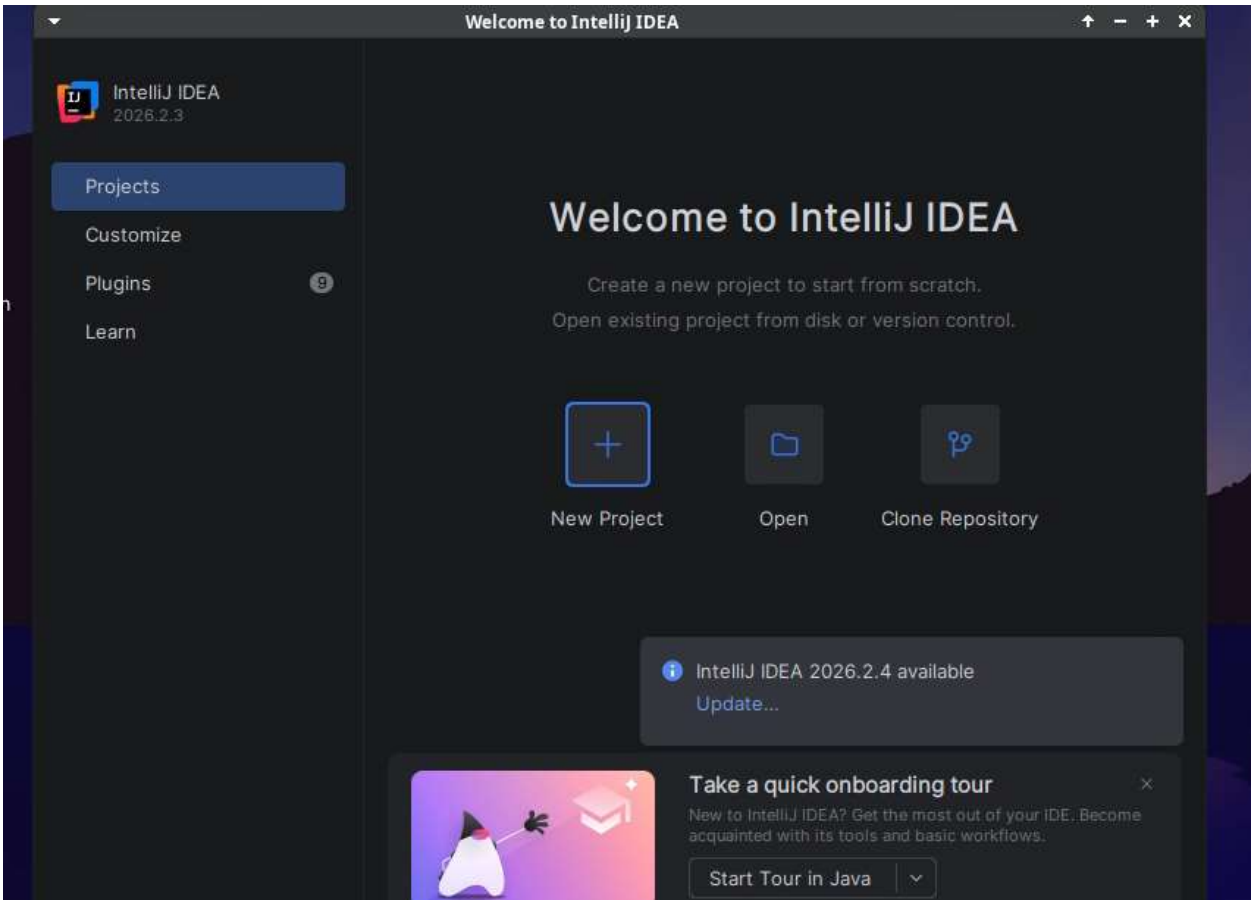
```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ sudo docker build -t lab5:1.0 .
[+] Building 4.8s (1/2)                                                  docker:default
 => [internal] load metadata for docker.io/library/node:13-alpine                 3.8s
```

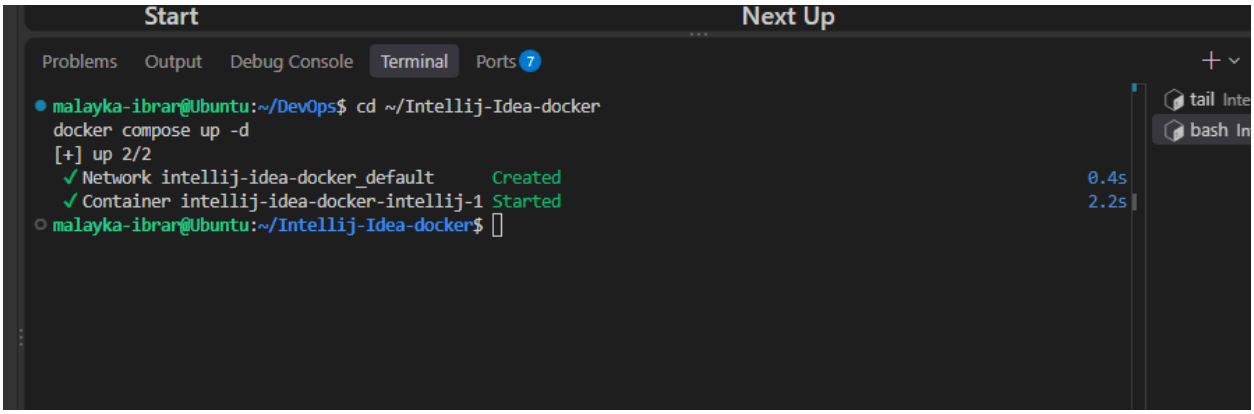
```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ sudo docker images
                                                  i Info →   U  In Use
IMAGE                                     ID             DISK USAGE   CONTENT SIZE   EXTRA
intellij-idea-docker-intellij:latest      bbcac1f94027         15GB          4.68GB    U
lab5:1.0                                  331df1c5580a        417MB          85.2MB
linuxserver/webtop:ubuntu-xfce            432475b977d9        5.15GB          1.32GB
lscr.io/linuxserver/webtop:ubuntu-xfce    432475b977d9        5.15GB          1.32GB
sonatype/nexus3:latest                    91a9af44af77        1.81GB           809MB    U
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ docker --version
Docker version 29.9.2, build a82e528
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/react-nodejs-example/api$ cd ~/IdeaProjects/node-project
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ ls -l Dockerfile
-rw-r--r-- 1 abc abc 119 Nov 11 20:45 Dockerfile
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$ cd ~/IdeaProjects
malayka-ibrar@ubuntu:~/IdeaProjects$ git clone https://gitlab.com/nanuchi/java-app.git
Cloning into 'java-app'...
remote: Enumerating objects: 222, done.
remote: Counting objects: 100% (112/112), done.
remote: Compressing objects: 100% (59/59), done.
remote: Total 222 (delta 33), reused 86 (delta 17), pack-reused 110 (from 1)
Receiving objects: 100% (222/222), 15.55 MiB | 301.00 KiB/s, done.
Resolving deltas: 100% (53/53), done.
malayka-ibrar@ubuntu:~/IdeaProjects$
```


Welcome to IntelliJ IDEA
IntelliJ IDEA
2026.2.3
Projects
Customize
Plugins
9
Learn
Welcome to IntelliJ IDEA
Create a new project to start from scratch.
Open existing project from disk or version control.
New Project
Open
Clone Repository
IntelliJ IDEA 2026.2.4 available
Update...
Take a quick onboarding tour
New to IntelliJ IDEA? Get the most out of your IDE. Become acquainted with its tools and basic workflows.
Start Tour in Java


Start
Next Up
Problems
Output
Debug Console
Terminal
Ports 7
malayka-ibrar@Ubuntu:~/DevOps$ cd ~/Intellij-Idea-docker
docker compose up -d
[+] up 2/2
✓ Network intellij-idea-docker_default Created 0.4s
✓ Container intellij-idea-docker-intellij-1 Started 2.2s
malayka-ibrar@Ubuntu:~/Intellij-Idea-docker$
tail
bash

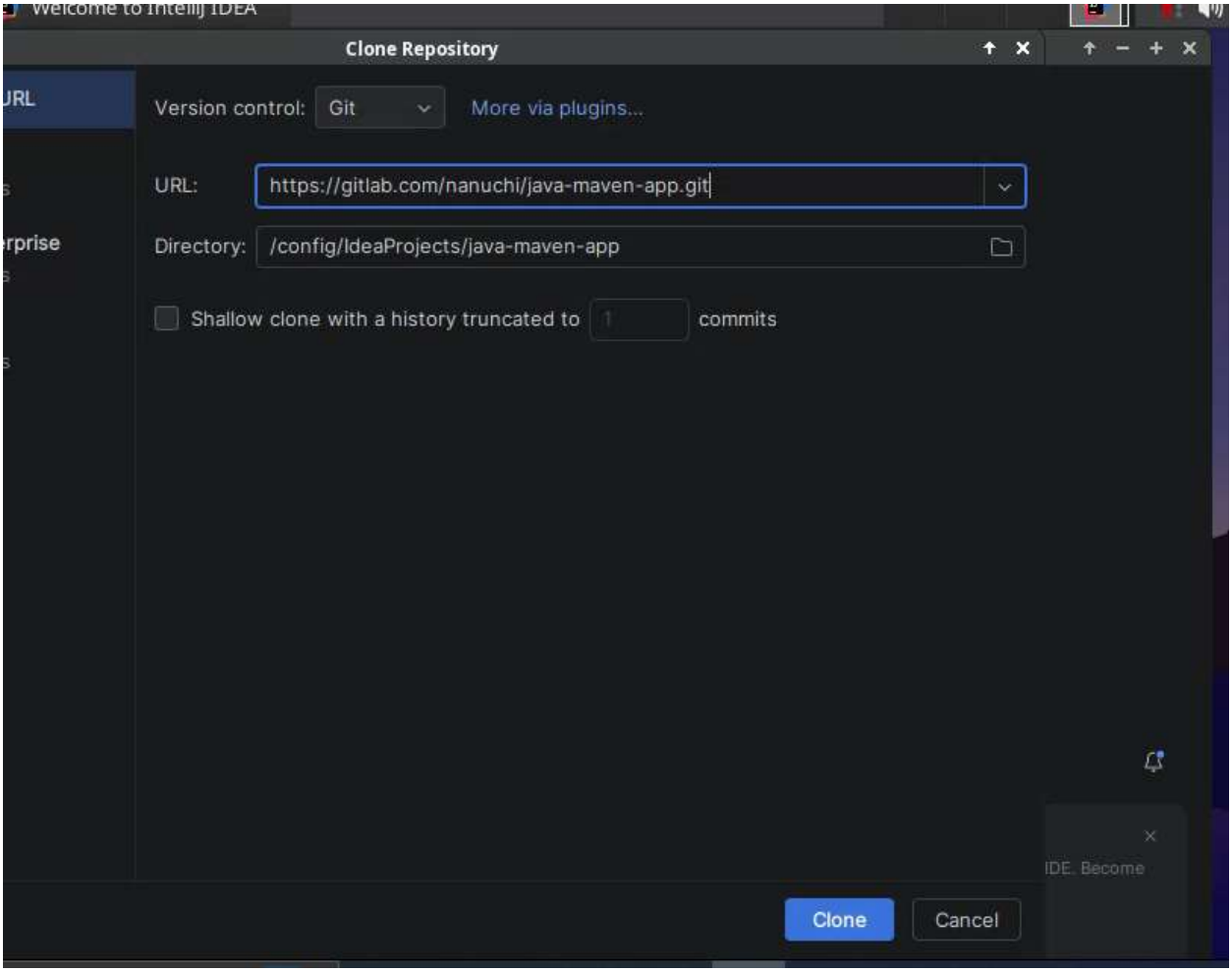

[INFO] --
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$ ls -lh target/
total 17M
drwxr-xr-x 4 abc abc 4.0K Nov 10 17:40 classes
drwxr-xr-x 3 abc abc 4.0K Nov 10 17:40 generated-sources
-rw-r--r-- 1 abc abc 17M Nov 10 17:48 java-maven-app-1.1.0-SNAPSHOT.jar
-rw-r--r-- 1 abc abc 3.0K Nov 10 17:48 java-maven-app-1.1.0-SNAPSHOT.jar.original
drwxr-xr-x 2 abc abc 4.0K Nov 10 17:41 maven-archiver
drwxr-xr-x 3 abc abc 4.0K Nov 10 17:40 maven-status
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$

```
) in auto mode
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$ mvn --version
Apache Maven 3.9.12
Maven home: /usr/share/maven
Java version: 21.0.13.0.3, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk
-amd64
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-38-generic", arch: "amd64", family: "unix"
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$
```

```
[INFO] ------------------------------------------------------------------------
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$ ls -lh target/
total 17M
drwxr-xr-x 4 abc abc 4.0K Nov 10 17:40 classes
drwxr-xr-x 3 abc abc 4.0K Nov 10 17:40 generated-sources
-rw-r--r-- 1 abc abc  17M Nov 10 17:48 java-maven-app-1.1.0-SNAPSHOT.jar
-rw-r--r-- 1 abc abc 3.0K Nov 10 17:48 java-maven-app-1.1.0-SNAPSHOT.jar.original
drwxr-xr-x 2 abc abc 4.0K Nov 10 17:41 maven-archiver
drwxr-xr-x 3 abc abc 4.0K Nov 10 17:40 maven-status
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$
```

```
Downloaded from central: https://repo.maven.apache.org/maven2/org/eclipse/sisu/org.eclipse.sisu.inject/0.3.4/o
.sisu.inject-0.3.4.jar (379 kB at 70 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.2.1/plexus-ut
jar (262 kB at 47 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/net/java/dev/jna/jna-platform/5.5.0/jna-platform
 (2.7 MB at 221 kB/s)
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  02:08 min
[INFO] Finished at: 2026-11-12T14:41:33+05:00
[INFO] ------------------------------------------------------------------------
root@ubuntu:~/IdeaProjects/java-maven-app#
```

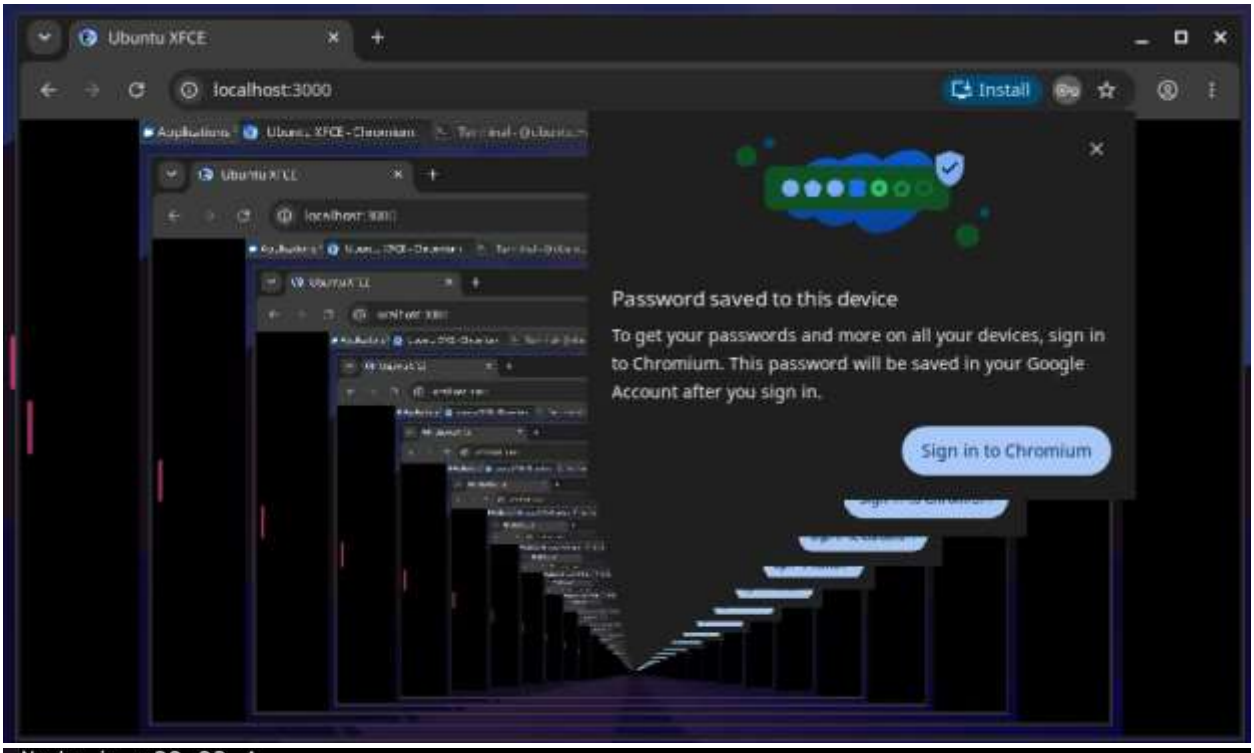





```
malayka-ibrar@ubuntu:~/IdeaProjects/java-maven-app$ java -jar target/java-maven-app-1.1.0-SNAPSHOT.j
ar

  .   ____          _            __ _ _
 /\\ / ___'_ __ _ _(_)_ __  __ _ \ \ \ \
( ( )\___ | '_ | '_| | '_ \/ _` | \ \ \ \
 \\/  ___)| |_)| | | | | || (_| |  ) ) ) )
  '  |____| .__|_| |_|_| |_\__, | / / / /
 =========|_|==============|___/=/_/_/_/
 :: Spring Boot ::        (v2.3.4.RELEASE)

2026-11-10 17:52:08.381  INFO 2943 --- [           main] com.example.Application                  :
Starting Application on ubuntu with PID 2943 (/config/IdeaProjects/java-maven-app/target/java-maven-
app-1.1.0-SNAPSHOT.jar started by abc in /config/IdeaProjects/java-maven-app)
2026-11-10 17:52:08.396  INFO 2943 --- [           main] com.example.Application                  :
No active profile set, falling back to default profiles: default
2026-11-10 17:52:11.133  INFO 2943 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  :
Tomcat initialized with port(s): 8080 (http)
2026-11-10 17:52:11.210  INFO 2943 --- [           main] o.apache.catalina.core.StandardService   :
Starting service [Tomcat]
2026-11-10 17:52:11.212  INFO 2943 --- [           main] org.apache.catalina.core.StandardEngine  :
Starting Servlet engine: [Apache Tomcat/9.0.38]
2026-11-10 17:52:11.368  INFO 2943 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       :
Initializing Spring embedded WebApplicationContext
2026-11-10 17:52:11.370  INFO 2943 --- [           main] w.s.c.ServletWebServerApplicationContext :
Root WebApplicationContext: initialization completed in 2704 ms
2026-11-10 17:52:11.523  INFO 2943 --- [           main] com.example.Application                  :
Java app started
```




```
Node.js v22.23.4
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$ nano server.js
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$ npm start

> bootcamp-node-project@1.0.0 start
> node server.js

app listening on port 3000!
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$ cd ~/IdeaProjects/node-project/app
npm start

> bootcamp-node-project@1.0.0 start
> node server.js

app listening on port 3000!
^C
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ node -v
npm -v
v22.23.4
10.9.9
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/java-app$ ^C
malayka-ibrar@ubuntu:~/IdeaProjects/java-app$ ^C
malayka-ibrar@ubuntu:~/IdeaProjects/java-app$ cd ~/IdeaProjects
git clone https://gitlab.com/devopsbootcamp3/node-project.git
Cloning into 'node-project'...
remote: Enumerating objects: 54, done.
remote: Total 54 (delta 0), reused 0 (delta 0), pack-reused 54 (from 1)
Receiving objects: 100% (54/54), 180.57 KiB | 233.00 KiB/s, done.
Resolving deltas: 100% (20/20), done.
malayka-ibrar@ubuntu:~/IdeaProjects$ cd ~/IdeaProjects/node-project
ls
Dockerfile  Jenkinsfile  README.md  app
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project$ cd ~/IdeaProjects/node-project/app
npm install
npm warn old lockfile
npm warn old lockfile The package-lock.json file was created with an old version of npm,
npm warn old lockfile so supplemental metadata must be fetched from the registry.
npm warn old lockfile
npm warn old lockfile This is a one-time fix-up, please be patient...
npm warn old lockfile
npm warn deprecated whatwg-encoding@1.0.5: Use @exodus/bytes instead for a more spec-conformant a
nd faster implementation
npm warn deprecated urix@0.1.0: Please see https://github.com/lydell/urix#deprecated
npm warn deprecated w3c-hr-time@1.0.2: Use your platform's native performance.now() and performan
ce.timeOrigin.
npm warn deprecated uuid@8.3.2: uuid@10 and below is no longer supported.  For ESM codebases, upd
ate to uuid@latest.  For CommonJS codebases, use uuid@11 (but be aware this version will likely b
e deprecated in 2028).
npm warn deprecated source-map-url@0.4.1: See https://github.com/lydell/source-map-url#deprecated
npm warn deprecated source-map-resolve@0.5.3: See https://github.com/lydell/source-map-resolve#de
precated
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated resolve-url@0.2.1: https://github.com/lydell/resolve-url#deprecated
npm warn deprecated sane@4.1.0: some dependency vulnerabilities fixed, support for node < 10 drop
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$ npm pack
npm notice
npm notice 📦  bootcamp-node-project@1.0.0
npm notice Tarball Contents
npm notice 60.5kB images/profile-andrea.jpg
npm notice 17.5kB images/profile-ari.jpeg
npm notice 1.2kB index.html
npm notice 283B package.json
npm notice 724B server.js
npm notice 205B server.test.js
npm notice Tarball Details
npm notice name: bootcamp-node-project
npm notice version: 1.0.0
npm notice filename: bootcamp-node-project-1.0.0.tgz
npm notice package size: 79.4 kB
npm notice unpacked size: 80.4 kB
npm notice shasum: ed73d22d636ef0dfcd62438fdd15427a920a4b08
npm notice integrity: sha512-wJvZ5wvMZAKZW[...]s1h4BOA8XdXew==
npm notice total files: 6
npm notice
bootcamp-node-project-1.0.0.tgz
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$ npm test

> bootcamp-node-project@1.0.0 test
> jest

Browserslist: caniuse-lite is outdated. Please run:
npx browserslist@latest --update-db

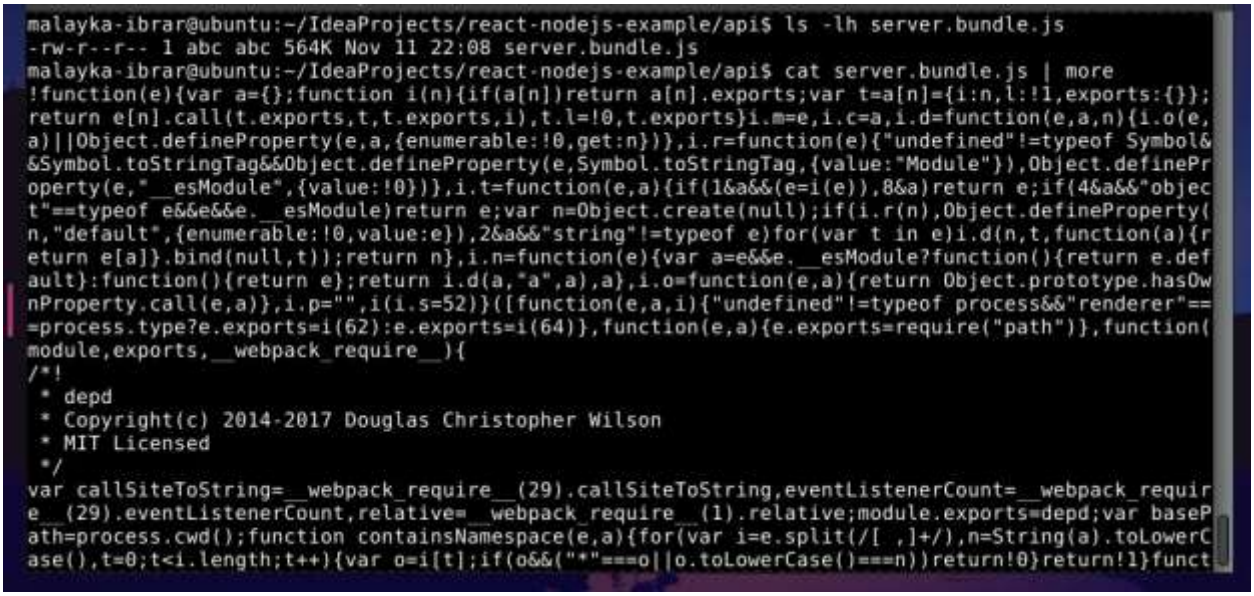
Why you should do it regularly:
https://github.com/browserslist/browserslist#browsers-data-updating
 PASS  ./server.test.js
  ✓ main index.html file exists (10 ms)

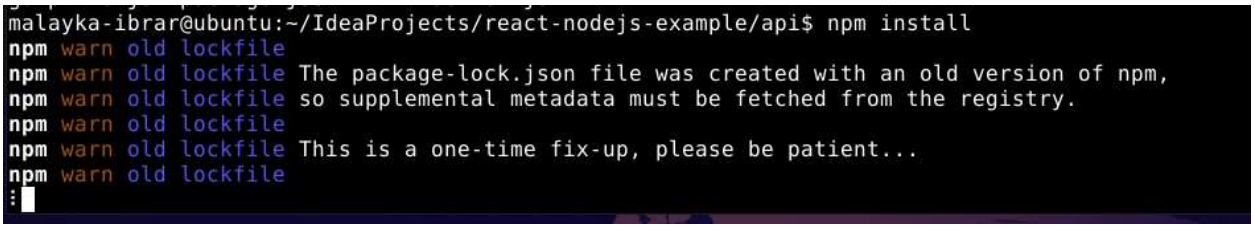
Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   0 total
Time:        7.174 s
Ran all test suites.
malayka-ibrar@ubuntu:~/IdeaProjects/node-project/app$
```

```
malayka-ibrar@ubuntu:~/IdeaProjects/react-nodejs-example/api$ NODE_OPTIONS=--openssl-legacy-provi
der npm run build

> react-nodejs-example@1.0.0 build
> webpack

Hash: abef9fc2cbaaf24f1946
Version: webpack 4.43.0
Time: 6391ms
Built at: 11/11/2026 10:08:06 PM
          Asset     Size  Chunks             Chunk Names
server.bundle.js  563 KiB       0  [emitted]  main
Entrypoint main = server.bundle.js
  [1] external "path" 42 bytes {0} [built]
  [4] external "fs" 42 bytes {0} [built]
  [6] external "buffer" 42 bytes {0} [built]
  [9] external "stream" 42 bytes {0} [built]
 [20] external "http" 42 bytes {0} [built]
 [22] external "util" 42 bytes {0} [built]
 [30] external "events" 42 bytes {0} [built]
 [32] external "net" 42 bytes {0} [built]
 [40] external "querystring" 42 bytes {0} [built]
```


malayka-ibrar@ubuntu:~/IdeaProjects/react-nodejs-example/api$ ls -lh server.bundle.js
-rw-r--r-- 1 abc abc 564K Nov 11 22:08 server.bundle.js
malayka-ibrar@ubuntu:~/IdeaProjects/react-nodejs-example/api$ cat server.bundle.js | more
!function(e){var a={};function i(n){if(a[n])return a[n].exports;var t=a[n]={i:n,l:!1,exports:{}};
return e[n].call(t.exports,t,t.exports,i),t.l=!0,t.exports}i.m=e,i.c=a,i.d=function(e,a,n){i.o(e,
a)||Object.defineProperty(e,a,{enumerable:!0,get:n})},i.r=function(e){"undefined"!=typeof Symbol&
&Symbol.toStringTag&&Object.defineProperty(e,Symbol.toStringTag,{value:"Module"}),Object.definePr
operty(e,"__esModule",{value:!0})},i.t=function(e,a){if(1&a&&(e=i(e)),8&a)return e;if(4&a&&"objec
t"==typeof e&&e&&e.__esModule)return e;var n=Object.create(null);if(i.r(n),Object.defineProperty(
n,"default",{enumerable:!0,value:e}),2&a&&"string"!=typeof e)for(var t in e)i.d(n,t,function(a){r
eturn e[a]}.bind(null,t));return n},i.n=function(e){var a=e&&e.__esModule?function(){return e.def
ault}:function(){return e};return i.d(a,"a",a),a},i.o=function(e,a){return Object.prototype.hasOw
nProperty.call(e,a)},i.p="",i(i.s=52)}([function(e,a,i){"undefined"!=typeof process&&"renderer"==
=process.type?e.exports=i(62):e.exports=i(64)},function(e,a){e.exports=require("path")},function(
module,exports,__webpack_require__){
/*!
 * depd
 * Copyright(c) 2014-2017 Douglas Christopher Wilson
 * MIT Licensed
 */
var callSiteToString=__webpack_require__(29).callSiteToString,eventListenerCount=__webpack_requir
e__(29).eventListenerCount,relative=__webpack_require__(1).relative;module.exports=depd;var baseP
ath=process.cwd();function containsNamespace(e,a){for(var i=e.split(/[,]+/),n=String(a).toLowerC
ase(),t=0;t<i.length;t++){var o=i[t];if(o&&("*"===o||o.toLowerCase()===n))return!0}return!1}funct


malayka-ibrar@ubuntu:~/IdeaProjects/react-nodejs-example/api$ npm install
npm warn old lockfile
npm warn old lockfile The package-lock.json file was created with an old version of npm,
npm warn old lockfile so supplemental metadata must be fetched from the registry.
npm warn old lockfile
npm warn old lockfile This is a one-time fix-up, please be patient...
npm warn old lockfile